Exploring Text

Descriptive Approaches

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Why Explore Before Modelling?

Key idea

Just as we compute descriptive statistics before running regressions, we should **explore text** before applying complex models.

- **Understand your data** — identify patterns, outliers, and data-quality issues early
- **Catch preprocessing errors** — missing documents, encoding problems, unexpected languages
- **Generate hypotheses** — exploratory analysis reveals structure that informs modelling choices

Descriptive text analysis is valuable in its own right — not merely a stepping stone to fancier methods.

Word Frequencies

What they show:

- Which terms dominate a corpus
- Differences across groups or time periods
- Raw counts or relative frequencies

Zipf's law: a small number of words account for most tokens; most words are rare.

Limitations

High-frequency words are often **stopwords**. Stemming and lemmatisation choices affect counts.

Example: Top 5 terms

Term	Frequency
government	1,842
member	1,536
minister	1,204
people	987
policy	763

After stopword removal; UK parliamentary debates.

Word Clouds

Strengths

- 👁 Intuitive and visually engaging
- Quick overview of dominant terms
- Useful for presentations to non-technical audiences

Limitations

- ⚠ No information about **context**
- Word size is hard to compare precisely
- Layout is arbitrary — can mislead
- Not reproducible across runs (random placement)

Rule of thumb

Word clouds are for **communication**, not **analysis**. Always supplement with frequency tables or bar charts.

Keywords-in-Context (KWIC)

Definition

Concordance lines showing a **target word** with its surrounding context, revealing *how* terms are used — not just how often.

Example: “immigration” in parliamentary debates

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...controls on  immigration  are essential to...
...debate about  immigration  has become too...
...the impact of immigration  on public services...
...reduce illegal immigration  through cooperation...
```

ℹ KWIC reveals framing, sentiment, and collocational patterns that frequency counts alone miss.

Collocations and N-grams

Key idea

Words that **co-occur more than expected by chance** carry meaning beyond individual terms.

Bigrams Two-word sequences: *climate change*, *prime minister*

Trigrams Three-word sequences: *cost of living*, *United Nations Security*

Identifying meaningful collocations:

- Raw co-occurrence counts (biased towards frequent words)
- **Pointwise Mutual Information (PMI)** — how much more often words co-occur than expected under independence
- Chi-squared and log-likelihood tests

Lexical Diversity

Type–Token Ratio (TTR)

$\text{TTR} = \frac{\text{Number of unique types}}{\text{Total number of tokens}}$ — a higher TTR indicates a **richer, more varied vocabulary**.

Applications in social science:

- Comparing **vocabulary richness** across political speakers
- Tracking linguistic complexity over time
- Assessing readability of policy documents

Caveat

TTR is sensitive to **document length** — longer texts tend to have lower TTR. Use standardised measures (e.g., MATTR, moving-average TTR) when comparing texts of different lengths.

Next Up

Practical 2: Descriptive Text Analysis

- Compute and visualise word frequencies
- Generate KWIC concordance lines
- Identify collocations and n-grams
- Measure lexical diversity across speakers

Open the **Practical 2** notebook in Google Colab
(Python or R — your choice!)